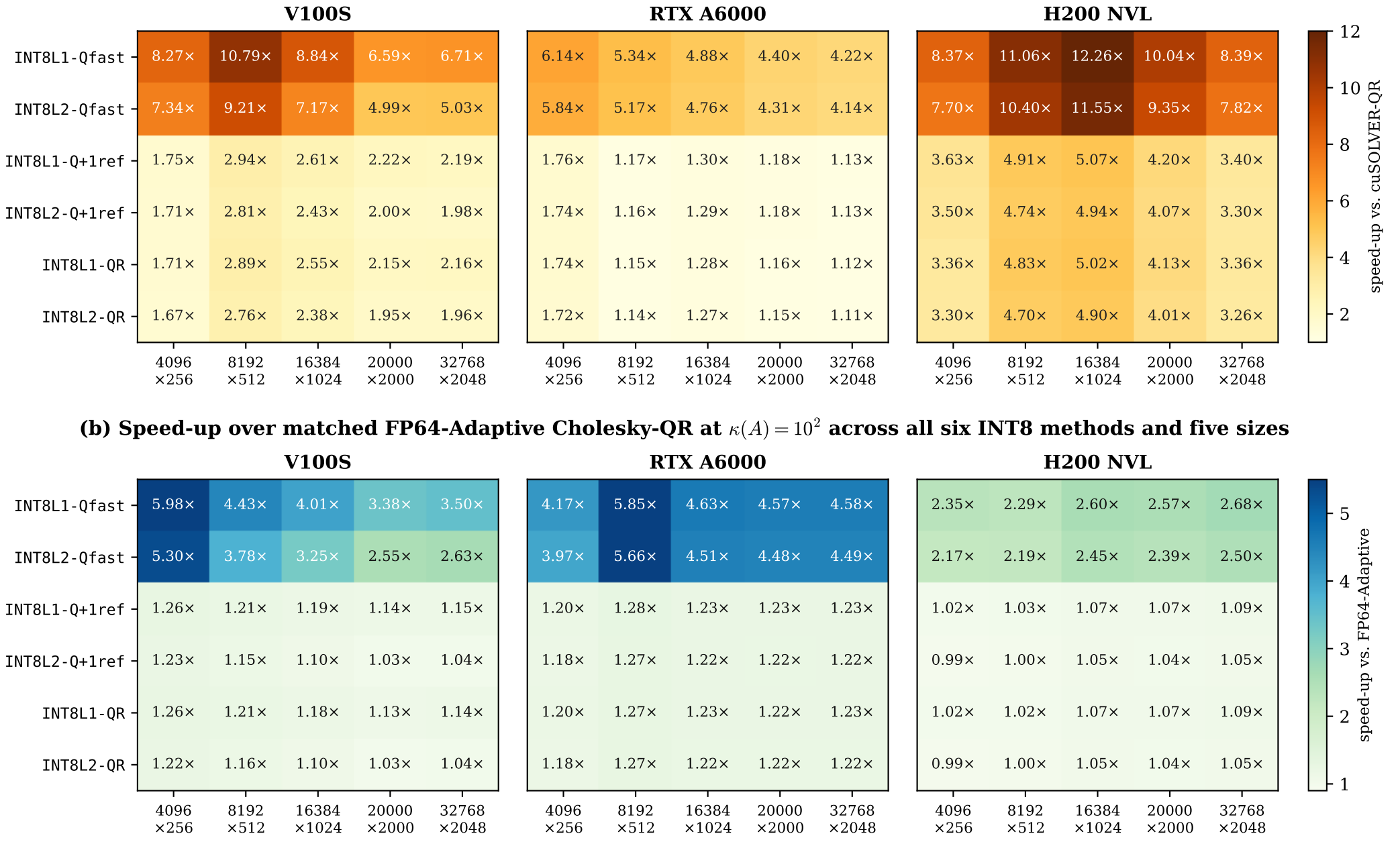
(a) Speed-up over vendor cuSOLVER-QR at $\kappa(A) = 1$ across all six INT8 methods and five sizes
V100S
RTX A6000
H200 NVL
INT8L1-Qfast
INT8L2-Qfast
INT8L1-Q+1ref
INT8L2-Q+1ref
INT8L1-QR
INT8L2-QR
8.27× 10.79× 8.84× 6.59× 6.71×
6.14× 5.34× 4.88× 4.40× 4.22×
8.37× 11.06× 12.26× 10.04× 8.39×
7.34× 9.21× 7.17× 4.99× 5.03×
5.84× 5.17× 4.76× 4.31× 4.14×
7.70× 10.40× 11.55× 9.35× 7.82×
1.75× 2.94× 2.61× 2.22× 2.19×
1.76× 1.17× 1.30× 1.18× 1.13×
3.63× 4.91× 5.07× 4.20× 3.40×
1.71× 2.81× 2.43× 2.00× 1.98×
1.74× 1.16× 1.29× 1.18× 1.13×
3.50× 4.74× 4.94× 4.07× 3.30×
1.71× 2.89× 2.55× 2.15× 2.16×
1.74× 1.15× 1.28× 1.16× 1.12×
3.36× 4.83× 5.02× 4.13× 3.36×
1.67× 2.76× 2.38× 1.95× 1.96×
1.72× 1.14× 1.27× 1.15× 1.11×
3.30× 4.70× 4.90× 4.01× 3.26×
4096 ×256
8192 ×512
16384 ×1024
20000 ×2000
32768 ×2048
12
10
8
6
4
2
speed-up vs. cuSOLVER-QR
(b) Speed-up over matched FP64-Adaptive Cholesky-QR at $\kappa(A) = 10^2$ across all six INT8 methods and five sizes
V100S
RTX A6000
H200 NVL
INT8L1-Qfast
INT8L2-Qfast
INT8L1-Q+1ref
INT8L2-Q+1ref
INT8L1-QR
INT8L2-QR
5.98× 4.43× 4.01× 3.38× 3.50×
4.17× 5.85× 4.63× 4.57× 4.58×
2.35× 2.29× 2.60× 2.57× 2.68×
5.30× 3.78× 3.25× 2.55× 2.63×
3.97× 5.66× 4.51× 4.48× 4.49×
2.17× 2.19× 2.45× 2.39× 2.50×
1.26× 1.21× 1.19× 1.14× 1.15×
1.20× 1.28× 1.23× 1.23× 1.23×
1.02× 1.03× 1.07× 1.07× 1.09×
1.23× 1.15× 1.10× 1.03× 1.04×
1.18× 1.27× 1.22× 1.22× 1.22×
0.99× 1.00× 1.05× 1.04× 1.05×
1.26× 1.21× 1.18× 1.13× 1.14×
1.20× 1.27× 1.23× 1.22× 1.23×
1.02× 1.02× 1.07× 1.07× 1.09×
1.22× 1.16× 1.10× 1.03× 1.04×
1.18× 1.27× 1.22× 1.22× 1.22×
0.99× 1.00× 1.05× 1.04× 1.05×
4096 ×256
8192 ×512
16384 ×1024
20000 ×2000
32768 ×2048
5
4
3
2
1
speed-up vs. FP64-Adaptive